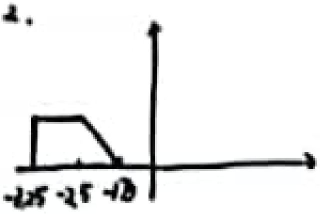
参考答案

1. $\delta(t) - e^{-2t}\varepsilon(t)$

2.



$F_1(j\omega) = \frac{1}{2} e^{3j\omega} F(j\omega)$

3. $\frac{(t+3)^2}{2}\varepsilon(t+3) - \frac{(t+2)^2}{2}\varepsilon(t+2) - \frac{(t+1)^2}{2}\varepsilon(t+1) + \frac{(t-1)^2}{2}\varepsilon(t-1) + \frac{(t-2)^2}{2}\varepsilon(t-2) + \frac{(t-3)^2}{2}\varepsilon(t-3)$

5 $\frac{12}{j\omega+5} - \frac{3}{j\omega-3}$

$\frac{12}{s+5}$ $Re(s) > -5$

6 不失真，部分失真

7 $r(t) = 4 + 1.5\cos(0.5t - \frac{3\pi}{4}) + 0.5\cos(1.5t - \frac{5\pi}{4})$

8 $r(t) = (9e^{-2t} - \frac{13}{2}e^{-13t} - \frac{1}{2}e^{-t})\varepsilon(t)$

9 (1) 0.5

(2) 载波：5×10^5 W

边频：4.25×10^4 W

平均：5.425×10^5 W

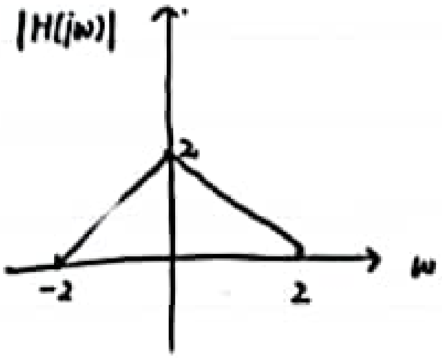
峰值：1.125×10^6 W

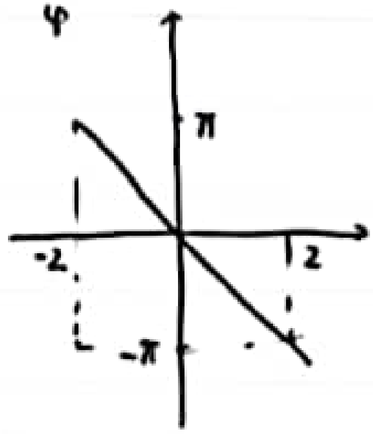
(3) 能耗大，效率低 / 降低直流分量，防止过调幅

10 (1) $\omega_0 = 1000$ $4 < \omega_1 < 1996$

$h(t) = \frac{\omega_1}{\pi} Sa[\omega_1(t-1)]$

7.





$e(t)=2+2\cos 0.5t\sin t$，求$r(t)$

8 $H(s)=\dfrac{s}{s^2+5s+6}$ $r_{zi}(0)=2$ $r_{zi}'(t)=1$ $e(t)=e^{-t}\varepsilon(t)$

求 $r(t)$ 并指出自然响应，受迫响应，瞬态响应，稳态响应

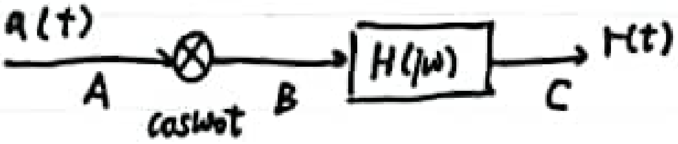
9 $a(t)=(1000+100\cos 2t+400\cos 4t)\cos 1000t$

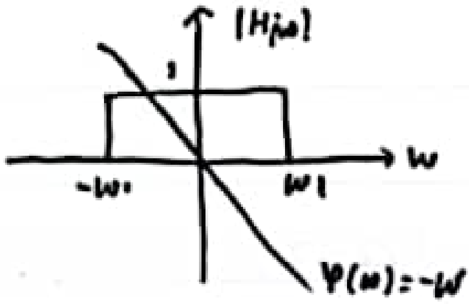
(1) 求调幅系数

(2) 求峰值功率，平均功率，边频功率，载波功率

(3) 试问该调幅有什么缺点，怎么改进

10





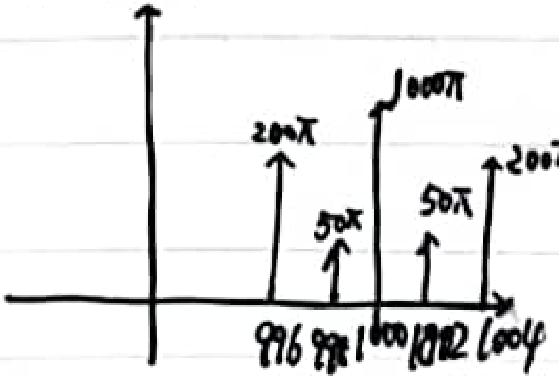
ω_0

(1) 求ω_0，ω_1，冲激响应

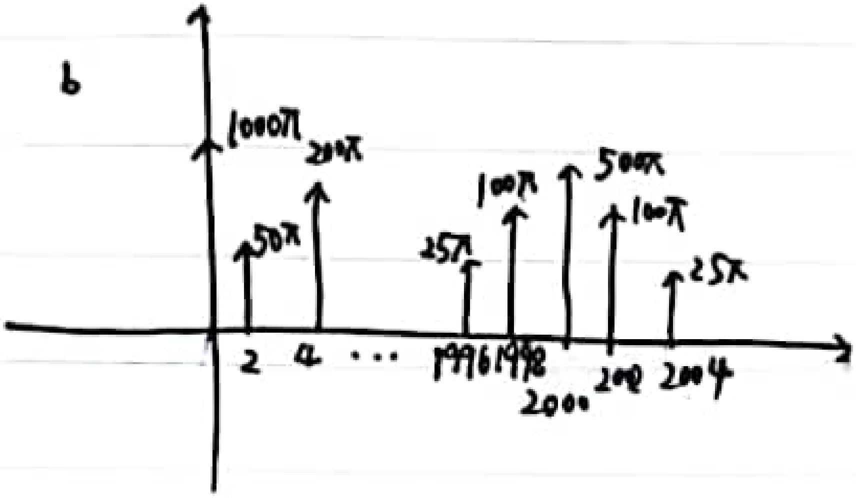
(2) 画出A，B，C三点频谱

(3) 求$r(t)$

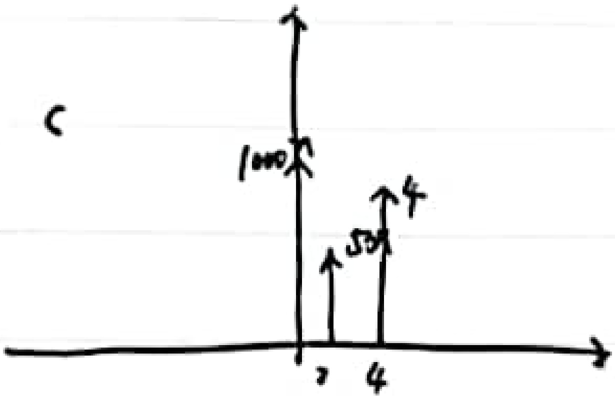
a （注：应该是双边 ）



b



c



(3) $r(t)=500+50\cos[2(t-1)]+200\cos[4(t-1)]$